

ENGINEERING CONTROL PHILOSOPHY

# Water-Tank PCS and SIS Control System

Split-range tank level control, duty/standby discharge pumps and independent 2oo3 high-high/low-low protection

| Document          | Revision | Issue date  | Status                                            |
|-------------------|----------|-------------|---------------------------------------------------|
| WT-PCS-SIS-CP-001 | A        | 02 Aug 2026 | Simulation Complete - Hardware Validation Pending |

**Prepared by Simon Kamua**

PLC/SIS Technician | ICSS Commissioning, Troubleshooting and Process Safety

## Document status and safety boundary

This document records an independently developed educational control-system design. It uses simulated values and sanitized tags and contains no employer-owned drawings, operating data, screenshots, controller backups or configuration files.

| Evidence label         | Meaning                                                                                                 |
|------------------------|---------------------------------------------------------------------------------------------------------|
| Design documented      | Process purpose, architecture, I/O, alarms, sequences and protection responses are defined.             |
| Desktop tested         | Reference-model checks and the interactive browser simulation have been exercised.                      |
| Not hardware validated | No compiled Omron project, field I/O, FAT, SAT, live process connection or SIL verification is claimed. |

**Safety notice: The ESP32-S3 is not a certified safety PLC. The demonstrator must not be used for an operational safety instrumented function.**

## 1. Purpose and scope

A reservoir feeds TK-101 through fail-close LCV-101. Two VFD pumps, P-101A and P-101B, discharge the tank in duty/standby service. The Omron PCS maintains level using an inventory-reduction controller whose split-range outputs progressively close the inlet valve and increase discharge-pump speed as level rises. A separate ESP32-S3 educational demonstrator receives LT-201A/B/C and applies independent 2oo3 votes: high-high closes and inhibits the inlet; low-low trips and inhibits both discharge pumps.

### 1.1 Objectives

- Maintain TK-101 close to the illustrative 55% operating setpoint under changing inlet and downstream conditions.
- Coordinate fail-close LCV-101 and the selected discharge pump through an explicit split-range characteristic.
- Select duty and standby pumps in the PCS and perform a status-verified bumpless changeover.
- Detect start failure, unexpected stop, VFD/motor faults, speed deviation, no-flow operation and loss of both pumps.
- Demonstrate separate 2oo3 high-high inlet shutdown and low-low pump shutdown functions with latching and manual reset.

### 1.2 Design assumptions

- LCV-101 is installed on the reservoir-to-tank inlet and fails closed on loss of its protective permit.

- P-101A/B are parallel discharge pumps downstream of TK-101, with a 30% illustrative minimum stable VFD speed.
- LT-101 is the normal PCS measurement and uses detected upscale burnout handling.
- LT-201A/B/C are separate protection measurements with individual quality and disagreement diagnostics.
- Protection outputs override PCS commands through independent de-energize-to-trip interfaces.

### 1.3 Explicit exclusions

- No SIL claim, SIL verification, PFDavg calculation, certified diagnostics or proof-test coverage is provided.
- No live process connection, hydraulic sizing, NPSH study, valve Cv calculation, pump-curve validation or network design is included.
- Illustrative setpoints, tuning, delays and actuator limits are not suitable for direct plant application.
- The ESP32-S3 is not a certified safety PLC and must not be used for operational protection.

## 2. Process and control architecture

The normal PCS and the educational protection layer are separated by measurement and function. LT-101 supports continuous PCS control. LT-201A/B/C support separate high-high and low-low votes. Protective outputs override ordinary PCS commands.

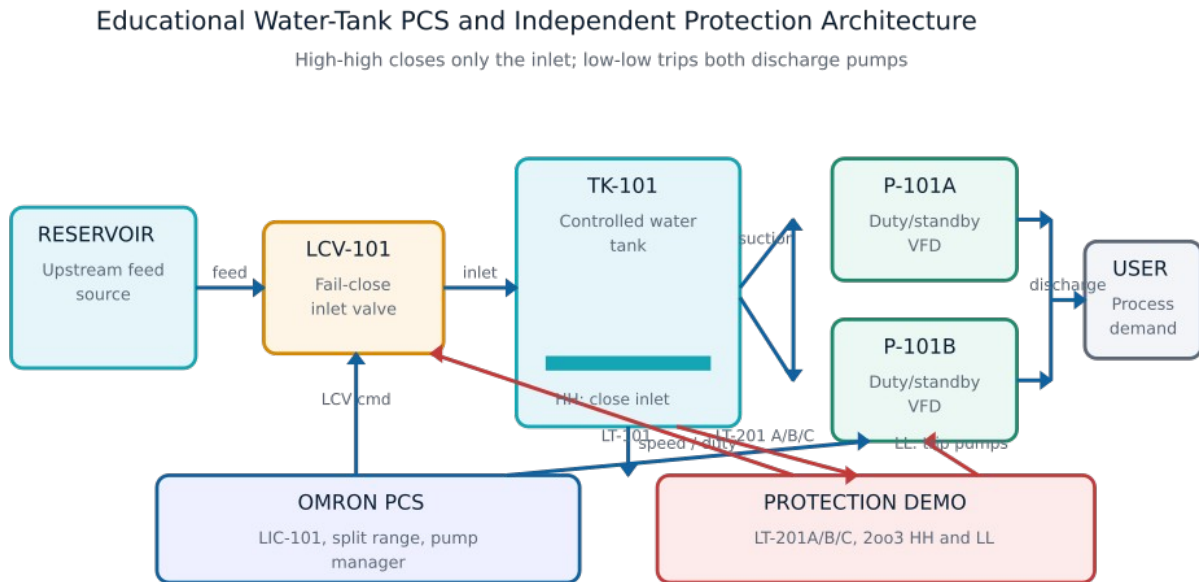


Figure 1 - Conceptual process and control architecture

### 2.1 Functional separation

| Layer                 | Normal responsibility                                                                               | Protective override                                                                |
|-----------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Omron PCS             | Level control, split range, pump selection, permissives, changeover, alarms and operator interface. | Must accept HH inlet-trip and LL pump-trip status; cannot defeat them.             |
| ESP32-S3 demonstrator | No normal control responsibility.                                                                   | 2oo3 HH closes inlet; 2oo3 LL trips pumps; separate latches and reset permissives. |
| Final elements        | LCV-101 and selected VFD pump follow PCS demands when permits are healthy.                          | De-energized protection permits force the defined action.                          |

## 3. Normal control philosophy

### 3.1 Controller action

LIC-101 produces an inventory-reduction demand. The demand rises when tank level rises above the 55% setpoint and falls when tank level falls below setpoint. This naming makes the direction explicit: a larger output asks the system to reduce stored inventory.

### 3.2 Illustrative split range

| Demand band | LCV-101 response | Active pump response | Control intent                                                     |
|-------------|------------------|----------------------|--------------------------------------------------------------------|
| 0-35%       | 100% open        | 30% minimum speed    | Recover low inventory while preserving minimum discharge.          |
| 35-65%      | 100% to 40% open | 30% to 50% speed     | Overlapping coordinated response around normal operation.          |
| 65-100%     | 40% to 0% open   | 50% to 100% speed    | Reduce high inventory by restricting inlet and increasing outflow. |

The overlap, limits, gains and ramps are simulation assumptions. Final values require pump curves, valve characteristics, process dynamics and commissioning tests.

### 3.3 PCS level-transmitter failure

LT-101 uses detected upscale burnout handling. A bad or out-of-range signal raises an alarm, forces the inlet closed and blocks new automatic pump starting. A running pump is held at a configured tracked bias pending operator intervention. Independent protection remains available through LT-201A/B/C.

## 4. Operating modes and sequence

| Mode/state     | Control intent                                                                                                 |
|----------------|----------------------------------------------------------------------------------------------------------------|
| Stopped        | Pumps stopped; inlet closed unless an authorized fill mode is active; monitoring and protection remain active. |
| Manual         | Authorized output proving with equipment interlocks and independent protection overrides retained.             |
| Automatic      | LIC-101 regulates tank inventory while the PCS manages the selected duty pump and standby availability.        |
| Changeover     | The incoming pump starts and ramps to tracked demand before the outgoing pump ramps down and stops.            |
| High-high trip | The protection layer closes and inhibits LCV-101; an eligible discharge pump remains available.                |
| Low-low trip   | The protection layer trips and inhibits both discharge pumps until level recovery and manual reset.            |
| Degraded       | A bad or disagreeing channel is annunciated; reset is blocked until voting health is restored.                 |

### 4.1 Sequence of operations

| Step | Phase                    | Required behavior                                                                                                         |
|------|--------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1    | Pre-start validation     | Confirm LT-101 valid, no low-low trip, selected pump available, downstream path healthy and protection status acceptable. |
| 2    | Start duty pump          | Start the selected discharge pump at 30%; prove run feedback and speed response before the timeout.                       |
| 3    | Enable automatic control | Track current outputs, then release LIC-101 to automatic inventory control.                                               |
| 4    | Split-range response     | As level rises, progressively close LCV-101 and increase discharge-pump speed; reverse the response as level falls.       |
| 5    | Controlled changeover    | Start standby, prove run feedback, ramp to tracked demand, unload the old duty pump, stop it and transfer duty.           |
| 6    | Pump-fault transfer      | Remove the failed pump, latch the initiating fault and transfer to the healthy standby when available.                    |
| 7    | High-high protection     | On 2oo3 at 90%, close and inhibit LCV-101; do not trip a healthy discharge pump solely for high level.                    |
| 8    | Low-low protection       | On 2oo3 at 15%, trip and inhibit both discharge pumps to prevent dry running.                                             |
| 9    | Reset and recovery       | Require HH below 85% or LL above 20%, healthy channels and a deliberate manual reset.                                     |

## 5. Duty/standby pump management

### 5.1 Availability permissives

- Local emergency stop healthy.
- VFD ready and communications healthy.
- No motor overload or active VFD fault.
- Remote operating mode selected.
- Tank level above the low-low reset threshold.
- No active maintenance inhibit.
- Low-low protection trip reset.
- Downstream flow path available.

## 5.2 Bumpless changeover

| Step | Action                                            | Confirmation                                    |
|------|---------------------------------------------------|-------------------------------------------------|
| 1    | Validate the selected standby pump.               | All availability permissives true.              |
| 2    | Issue start command at minimum stable speed.      | Run feedback before timeout.                    |
| 3    | Ramp incoming pump toward tracked demand.         | Speed feedback within 2% of demand.             |
| 4    | Ramp outgoing pump toward zero.                   | Incoming pump remains stable.                   |
| 5    | Stop outgoing pump and transfer duty designation. | Stopped feedback and new duty status confirmed. |

## 5.3 Fault response

| Fault                                 | Detection | Automatic response                                         |
|---------------------------------------|-----------|------------------------------------------------------------|
| Pump command with no run feedback     | 5 s       | Fail start; mark unavailable; start healthy standby        |
| Run feedback lost with command active | 0.5 s     | Remove command; transfer to healthy standby                |
| VFD fault or motor overload           | 0 s       | Stop affected pump; latch fault; transfer duty             |
| Speed deviation > 10%                 | 5 s       | Alarm; hold changeover; transfer if performance inadequate |
| Low discharge flow while running      | 5 s       | Stop affected pump; attempt standby transfer               |
| Both pumps unavailable                | 0 s       | High-priority alarm; close inlet as level rises            |

## 6. Independent protection philosophy

LT-201A, LT-201B and LT-201C are treated as independent educational input channels. A protective action requires two valid channels to cross the applicable threshold. A single abnormal channel does not trip, but disagreement or invalid quality is annunciated and reset is blocked until channel health returns.

| Initiating event      | Vote                  | Automatic action                                                             | Reset condition                                   |
|-----------------------|-----------------------|------------------------------------------------------------------------------|---------------------------------------------------|
| High-high level       | 2oo3 $\geq$ 90%       | Close and inhibit LCV-101. Keep an eligible discharge pump available.        | All valid channels < 85% and manual reset.        |
| Low-low level         | 2oo3 $\leq$ 15%       | Trip and inhibit P-101A and P-101B.                                          | All valid channels > 20% and manual reset.        |
| Channel disagreement  | Spread > 5% for delay | Degraded alarm; identify channel; no automatic trip from disagreement alone. | Agreement restored and acknowledged.              |
| Invalid channel       | Quality diagnostic    | Degraded alarm; invalid channel does not count as safe; block reset.         | All channels valid and voting state verified.     |
| Protection power loss | De-energize-to-trip   | Close inlet and remove pump permit.                                          | Power and diagnostics healthy, then manual reset. |

### 6.1 Real-SIS boundary

A real SIS requires an approved Safety Requirements Specification, hazard analysis, independence review, SIL verification, certified hardware, proof-test procedures and controlled management of bypasses and overrides. Those lifecycle activities are outside this portfolio demonstrator.

## 7. Setpoints and alarms

| Parameter             | Value | Units  | Purpose             |
|-----------------------|-------|--------|---------------------|
| Normal level setpoint | 55    | %      | PCS target          |
| High-level alarm      | 80    | %      | Operator warning    |
| High-high trip        | 90    | %      | 2oo3 inlet shutdown |
| High-high reset       | 85    | %      | Reset hysteresis    |
| Low-level alarm       | 25    | %      | Operator warning    |
| Low-low trip          | 15    | %      | 2oo3 pump shutdown  |
| Low-low reset         | 20    | %      | Reset hysteresis    |
| Channel disagreement  | 5     | % span | Degraded alarm      |

### 7.1 Alarm and interlock register

| ID    | Initiating condition               | Priority | Delay | Response                                                                  | Reset            |
|-------|------------------------------------|----------|-------|---------------------------------------------------------------------------|------------------|
| A-001 | LT-101 $\geq$ 80%                  | High     | 2 s   | Warn operator; controller increases inventory reduction                   | Auto clear       |
| A-002 | LT-101 $\leq$ 25%                  | High     | 2 s   | Warn operator; controller reduces discharge demand                        | Auto clear       |
| A-003 | LT-101 bad or out of range         | High     | 1 s   | Close inlet; block new auto start; hold running pump at safe tracked bias | Manual confirm   |
| A-004 | Any two LT-201 channels $\geq$ 90% | Trip     | 0.5 s | Close and inhibit LCV-101; keep eligible discharge pump available         | Manual below 85% |
| A-005 | Any two LT-201 channels $\leq$ 15% | Trip     | 0.5 s | Trip and inhibit P-101A/B                                                 | Manual above 20% |
| A-006 | Protection-channel spread > 5%     | Medium   | 5 s   | Identify disagreeing channel; degraded alarm                              | Healthy +        |



| ID    | Initiating condition                  | Priority | Delay | Response                                                   | Reset                  |
|-------|---------------------------------------|----------|-------|------------------------------------------------------------|------------------------|
|       |                                       |          |       |                                                            | acknowledge            |
| A-007 | Any protection channel invalid        | High     | 1 s   | Degraded alarm; block protective reset                     | All channels valid     |
| A-008 | Pump command with no run feedback     | High     | 5 s   | Fail start; mark unavailable; start healthy standby        | Manual / auto transfer |
| A-009 | Run feedback lost with command active | High     | 0.5 s | Remove command; transfer to healthy standby                | Manual                 |
| A-010 | VFD fault or motor overload           | High     | 0 s   | Stop affected pump; latch fault; transfer duty             | Manual                 |
| A-011 | Speed deviation > 10%                 | Medium   | 5 s   | Alarm; hold changeover; transfer if performance inadequate | Auto clear / manual    |
| A-012 | Low discharge flow while running      | High     | 5 s   | Stop affected pump; attempt standby transfer               | Manual                 |
| A-013 | Both pumps unavailable                | High     | 0 s   | High-priority alarm; close inlet as level rises            | Availability restored  |
| A-014 | LCV demand/position mismatch          | High     | 5 s   | Alarm; inhibit unsafe restart                              | Manual                 |

## 8. Functional I/O

The following signals are conceptual. Actual addresses, marshalling, electrical protection, network mapping and channel diagnostics remain hardware-project deliverables.

| Tag              | I/O    | Signal       | Description                    | Range/state             | Fail state            | Owner                  |
|------------------|--------|--------------|--------------------------------|-------------------------|-----------------------|------------------------|
| LT-101           | AI     | 4-20 mA      | PCS tank level                 | 0-100%                  | Upscale / bad quality | PCS                    |
| LT-201A          | AI     | 4-20 mA      | Protection level channel A     | 0-100%                  | Invalid channel       | Protection demo        |
| LT-201B          | AI     | 4-20 mA      | Protection level channel B     | 0-100%                  | Invalid channel       | Protection demo        |
| LT-201C          | AI     | 4-20 mA      | Protection level channel C     | 0-100%                  | Invalid channel       | Protection demo        |
| FIT-101          | AI     | 4-20 mA      | Common pump-discharge flow     | 0-150 m <sup>3</sup> /h | Bad quality           | PCS                    |
| PIT-101          | AI     | 4-20 mA      | Common pump-discharge pressure | 0-10 bar                | Bad quality           | PCS                    |
| LCV-101.POS      | AI     | 4-20 mA      | Inlet-valve position feedback  | 0-100%                  | Bad quality           | PCS                    |
| LCV-101.CMD      | AO     | 4-20 mA      | Inlet-valve position demand    | 0-100%                  | 0% / closed           | PCS + HH override      |
| P-101A.SPD       | AO/Net | 0-100%       | Pump A speed reference         | 0-100%                  | 0%                    | PCS                    |
| P-101B.SPD       | AO/Net | 0-100%       | Pump B speed reference         | 0-100%                  | 0%                    | PCS                    |
| P-101A.RUN       | DI     | 24 Vdc       | Pump A run feedback            | Run/Stop                | Stopped               | PCS                    |
| P-101B.RUN       | DI     | 24 Vdc       | Pump B run feedback            | Run/Stop                | Stopped               | PCS                    |
| P-101A.READY     | DI/Net | Boolean      | Pump A VFD ready               | Ready/Not ready         | Not ready             | PCS                    |
| P-101B.READY     | DI/Net | Boolean      | Pump B VFD ready               | Ready/Not ready         | Not ready             | PCS                    |
| P-101A.FLT       | DI/Net | Boolean      | Pump A VFD fault               | Healthy/Fault           | Fault                 | PCS                    |
| P-101B.FLT       | DI/Net | Boolean      | Pump B VFD fault               | Healthy/Fault           | Fault                 | PCS                    |
| P-101A.REM       | DI     | 24 Vdc       | Pump A remote mode             | Remote/Local            | Local                 | PCS                    |
| P-101B.REM       | DI     | 24 Vdc       | Pump B remote mode             | Remote/Local            | Local                 | PCS                    |
| P-101A.START     | DO     | 24 Vdc       | Pump A start/stop command      | Run/Stop                | Stop                  | PCS + LL override      |
| P-101B.START     | DO     | 24 Vdc       | Pump B start/stop command      | Run/Stop                | Stop                  | PCS + LL override      |
| SIS.RESET        | DI     | Momentary    | Protection manual reset        | Pressed/Released        | Released              | Protection demo        |
| SIS.INLET_PERMIT | DO     | De-energize  | Permit LCV-101 to open         | Permit/Trip             | Trip / close          | Protection demo        |
| SIS.PUMP_PERMIT  | DO     | De-energize  | Permit P-101A/B to run         | Permit/Trip             | Trip / stop           | Protection demo        |
| SIS.STATUS       | DI/Net | Boolean word | Trip/degraded status to PCS    | Status bits             | Unhealthy             | Protection demo to PCS |

## 9. Logic architecture

| Logic section           | Responsibility                                                                     |
|-------------------------|------------------------------------------------------------------------------------|
| Input conditioning      | Scaling, quality, range checks, debounce and conservative bad-signal substitution. |
| Modes and permissives   | Auto/manual, start/stop, selected duty, downstream readiness and trip status.      |
| Level control           | Inventory-reduction controller, tracking, limits and split-range mapping.          |
| Pump manager            | Availability, run proving, speed ramping, duty transfer and fault response.        |
| Protection interface    | Independent HH inlet override, LL pump override, status and reset interlocks.      |
| Protection demonstrator | Three channel diagnostics, 2oo3 HH/LL voting, separate latches and safe outputs.   |
| Alarms and events       | Priorities, first-out, changeover events and functional-test evidence.             |

### 9.1 Omron ladder-logic section schedule

| Rung    | Section              | Logic intent                                                       |
|---------|----------------------|--------------------------------------------------------------------|
| 000-019 | Input handling       | Scale LT/FIT/PIT; validate quality; debounce status                |
| 020-039 | Modes                | Auto/manual, start/stop and duty selection                         |
| 040-069 | Permissives          | Pump readiness, downstream path and protection status              |
| 070-109 | Control              | LIC inventory demand and split-range outputs                       |
| 110-169 | Pump manager         | Run proving, ramping and bumpless changeover                       |
| 170-219 | Faults               | Start fail, unexpected stop, speed mismatch, no flow and VFD trips |
| 220-249 | Protection interface | HH inlet override, LL pump override and status                     |
| 250-289 | Alarms               | Priorities, first-out and reset                                    |
| 290-319 | HMI                  | Commands, status, diagnostics and events                           |

## 10. Verification plan and desktop results

Desktop results demonstrate consistency between the design rules, deterministic checks and the public HTML simulation. They do not represent hardware FAT or SAT.

| Test   | Scenario                | Precondition            | Expected result                                                     | Result | Evidence               |
|--------|-------------------------|-------------------------|---------------------------------------------------------------------|--------|------------------------|
| WT-001 | Normal automatic start  | Level 55%; duty A ready | A starts at minimum speed; tracked auto control enabled             | PASS   | Reference model + HTML |
| WT-002 | High-level disturbance  | Auto stable             | LCV closes progressively; discharge speed increases                 | PASS   | Reference model + HTML |
| WT-003 | Low-level disturbance   | Auto stable above LL    | LCV opens; discharge speed reduces toward minimum                   | PASS   | Reference model + HTML |
| WT-004 | Duty changeover A to B  | Both pumps ready        | B reaches tracked demand before A stops                             | PASS   | HTML simulation        |
| WT-005 | Duty-pump failure       | A duty; B ready         | A is removed and B assumes duty                                     | PASS   | HTML fault injection   |
| WT-006 | PCS LT burnout          | Auto running            | Inlet closes; new auto starts blocked; protection remains available | PASS   | Reference model        |
| WT-007 | One LT-201 channel high | Other channels normal   | No HH trip; disagreement/degraded indication                        | PASS   | Reference model + HTML |
| WT-008 | Two channels high-high  | Protection healthy      | Inlet trip latches; discharge pump remains available                | PASS   | Reference model + HTML |
| WT-009 | One LT-201 channel low  | Other channels normal   | No LL trip                                                          | PASS   | Reference model        |
| WT-010 | Two channels low-low    | Protection healthy      | Both pumps trip and restart is inhibited                            | PASS   | Reference model + HTML |
| WT-011 | Reset with active cause | HH or LL remains active | Reset rejected                                                      | PASS   | Design                 |

| Test   | Scenario        | Precondition | Expected result               | Result | Evidence          |
|--------|-----------------|--------------|-------------------------------|--------|-------------------|
|        |                 |              |                               |        | inspection + HTML |
| WT-012 | Channel invalid | Trip latched | Degraded alarm; reset blocked | PASS   | Code inspection   |

## 10.1 Hardware FAT still required

- Compile the selected Omron project with no errors or unresolved I/O.
- Verify every analog scale, discrete state, fail direction and command/feedback pair.
- Test threshold boundaries, time delays, latches, reset permissives and controller restart behavior.
- Measure pump-transfer timing and confirm no unacceptable level or pressure disturbance.
- Record witness, defect, retest, backup and software-checksum evidence.

## 11. Engineering decisions and next development

| Decision                     | Engineering rationale                                                                                        |
|------------------------------|--------------------------------------------------------------------------------------------------------------|
| Inventory-reduction demand   | A rising controller output means more inventory removal: close the inlet and increase discharge capacity.    |
| Correct protective split     | High-high removes inflow but preserves healthy outflow; low-low removes pump outflow to prevent dry running. |
| Separate transmitters        | LT-101 serves normal control while LT-201A/B/C provide an independent educational voting layer.              |
| Manual reset with hysteresis | Separate reset thresholds prevent chatter and automatic restart at the trip boundary.                        |
| Bumpless pump transfer       | Proving and loading the incoming pump before unloading the outgoing pump reduces hydraulic disturbance.      |

### 11.1 Known limitations

- The simplified model omits pump curves, NPSH, valve stiction, line packing and pressure transients.
- PID tuning, split-range overlap and ramps are illustrative and require validated process data.
- The ST listing is a design reference, not a compiled Omron hardware project.
- The ESP32-S3 lacks certified diagnostics, systematic capability and independence evidence.

### 11.2 Recommended development

- Implement the selected Omron PID block with tracking, anti-windup and validated tuning.
- Add run-hour balancing, maintenance counters and equipment plausibility diagnostics.
- Build a low-voltage rig and execute witnessed FAT-style tests.
- Use certified hardware and complete an SRS and SIL verification before any genuine safety application.

End of controlled portfolio document.